Meta-SWAG: Theoretical Guarantees

Corrected Theorems and Proof Sketches

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Contents

1 Theoretical Guarantees

1.1 Variance Reduction Through Bayesian Model Averaging

Throughout this section we assume the following structural conditions on the checkpoint trajectory.

Assumption 1.1 (Evidence–variance alignment)**.** *For the retained checkpoints $\{\vartheta_k\}_{k=1}^K$, the weighting rule is monotone in an evidence score m_k, and higher evidence corresponds to lower checkpoint-level predictive variance:*

$$m_t \;\geq\; m_s \;\;\Longrightarrow\;\; V_t \;\leq\; V_s.$$

This is an idealized *assumption, not a generic property of arbitrary validation metrics.*

Structural assumptions (A1)–(A5) on the trajectory geometry and model class are stated separately and are taken as given throughout.

Theorem 1.1 (Idealized variance reduction under evidence–variance alignment)**.** *Let q_T be the Meta-SWAG posterior constructed from retained checkpoints satisfying Assumptions (A1)–(A5), and let $\bar{\pi}_{q_T}$ denote the posterior predictive policy. Under Assumption* **??***, the within-checkpoint component of the predictive variance satisfies*

$$\mathrm{Var}_{\mathrm{within}}(\bar{\pi}_{q_T}) \;\leq\; \mathrm{AM}(V_1,\ldots,V_K) \;=\; \mathrm{Var}_{\mathrm{within}}(\pi_{\mathrm{unif}}), \tag{1}$$

where V_k denotes checkpoint-level predictive variance and π_{unif} denotes the uniformly weighted checkpoint ensemble.

Moreover, when the weighting satisfies the stronger condition $w_k \propto V_k^{-1}$ (inverse-variance weighting), the bound tightens to the harmonic mean:

$$\mathrm{Var}_{\mathrm{within}}(\bar{\pi}_{q_T})\big|_{w_k \propto V_k^{-1}} \;=\; \mathrm{HM}(V_1,\ldots,V_K) \;\leq\; \mathrm{AM}(V_1,\ldots,V_K), \tag{2}$$

so that the ratio $\mathrm{HM}\,/\,\mathrm{AM} \leq 1$ (AM–HM inequality) quantifies the maximum gain from optimal inverse-variance weighting. If the between-checkpoint term remains controlled inside one stable region, this also yields a reduction relative to nearby point estimates.

Correction: The original statement claimed $\mathrm{Var}_{\mathrm{within}}(\bar{\pi}_{q_T}) \leq (\mathrm{HM}\,/\,\mathrm{AM})\,\mathrm{Var}_{\mathrm{within}}(\pi_{\mathrm{unif}})$ for any monotone evidence weighting. This is not generally true: the rearrangement inequality (or Chebyshev's sum inequality) applied to a weight sequence negatively correlated with $\{V_k\}$ yields only $\sum_k w_k V_k \leq \mathrm{AM}$, i.e. the bound in (**??**). The harmonic-mean bound (**??**) is a special case that holds with equality only under inverse-variance weights $w_k \propto V_k^{-1}$, a condition that must be stated explicitly.

Proof sketch. The within-checkpoint variance is $\mathrm{Var}_{\mathrm{within}}(\bar{\pi}_{q_T}) = \sum_{k=1}^K w_k V_k$. Under Assumption **??**, the weight sequence $\{w_k\}$ is *negatively* correlated with $\{V_k\}$ (heavier mass on lower-variance checkpoints). Chebyshev's sum inequality for oppositely ordered sequences then gives

$$\sum_k w_k V_k \;\leq\; \left(\sum_k w_k\right)\left(\frac{1}{K}\sum_k V_k\right) = \mathrm{AM}(V_1,\ldots,V_K),$$

establishing (**??**). For the harmonic-mean case, if $w_k = V_k^{-1} / \sum_j V_j^{-1}$, then

$$\sum_k w_k V_k = \frac{\sum_k V_k^{-1} \cdot V_k}{\sum_j V_j^{-1}} = \frac{K}{\sum_j V_j^{-1}} = \mathrm{HM}(V_1, \ldots, V_K),$$

giving (**??**) with equality. The AM–HM inequality HM $\leq$ AM for positive reals then confirms the chain $\mathrm{Var}_{\mathrm{within}}(\bar{\pi}_{q_T}) \leq \mathrm{HM} \leq \mathrm{AM} = \mathrm{Var}_{\mathrm{within}}(\pi_{\mathrm{unif}})$ in the inverse-variance case. Extending to a single point-estimate comparison requires the additional single-region control stated in the theorem. □

Corollary 1.1.1 (Alignment)**.** *Assume in addition that, on the retained LoRA posterior support, the best-of-n proxy–gold reward gap is locally upper-bounded by a monotone function of a predictive-dispersion functional that is itself controlled by the within-checkpoint predictive variance. Then any Meta-SWAG weighting rule that strictly reduces the within-checkpoint predictive variance also weakly reduces that local upper bound on reward over-optimisation relative to the corresponding point-estimate or uniform-ensemble baseline.*

Remark 1.1.1. Corollary **??** is a conditional statement: its conclusion is vacuous unless the bridge from predictive dispersion to the proxy–gold reward gap is established empirically. The significance of the theorem lies precisely in identifying this as the step the experimental protocol must validate rather than assume.

1.2 Posterior Covariance as Basin Geometry

Theorem 1.2 (Posterior covariance identifies local trajectory geometry)**.** *Suppose:*

(i) the retained trajectory remains in a stable local region of the adaptive optimisation process;

(ii) the dynamics admit a local linearisation in that region, with Jacobian J and Hessian H of the effective objective along the trajectory;

(iii) the retained checkpoint process is geometrically mixing; and

(iv) the leading eigenvalues of the local covariance are distinct *(eigenvalue-gap condition).*

Then, as $T \to \infty$, the Meta-SWAG covariance Σ_T converges to a matrix whose leading eigenspaces *agree with the dominant weakly-curved directions of the local trajectory geometry, while concentration in sharper directions scales with the local second-order structure of the effective objective. Without condition (iv), only convergence of the eigenspace as a whole (not of individual eigenvectors) can be guaranteed.*

Correction: The original statement omitted condition (iv). Convergence of individual eigenvectors requires distinct (or at least well-separated) leading eigenvalues; this is a standard requirement in perturbation theory for eigenvectors (Davis–Kahan theorem). The corrected statement either imposes an eigenvalue gap or weakens the conclusion to eigenspace convergence.

Proof sketch. Near a stable local region, the adaptive dynamics admit a local linearisation whose covariance is governed by J and H along the trajectory. The diagonal-plus-low-rank estimator of equation (7) is consistent for the dominant second-moment structure of this local process. Geometric mixing (condition (iii)) ensures that the checkpoint buffer estimates local covariance rather than transient drift between unrelated regions. The leading eigenvectors therefore align with the soft directions of the local orbit or basin, while narrow directions shrink under repeated averaging. Condition (iv) (eigenvalue gap) is required for convergence of individual eigenvectors; without it, one can only guarantee that the span of the leading eigenspace converges. □

Remark 1.2.1 (Diagnostic interpretation under subspace-sensitive shift). Assume additionally that deployment shift perturbs model behaviour primarily through the retained LoRA subspace, and that out-of-distribution loss increase is controlled by perturbation energy projected onto the leading eigenspaces of Σ_T. Under these conditions:

- a *spiked* spectrum of Σ_T is consistent with increased sensitivity to shifts aligned with the narrow adapter subspace;
- a *flatter* spectrum is consistent with more distributed local robustness.

This is an *interpretive remark*, not a mathematical corollary: it asserts consistency between spectral shape and qualitative behaviour, not a derived inequality. Any deployment-robustness interpretation requires an additional empirical assumption linking distribution shift to the local trajectory geometry summarised by Σ_T.

Correction: The original presented this as **Corollary 2**, implying it followed mathematically from Theorem 2. It does not: it states only *consistency*, not a derived bound. Relabelling as a Remark is both logically accurate and less likely to mislead a reader expecting a proved inequality.

1.3 Conceptual Limit: Self-Knowledge Bottlenecks Posterior Quality

The next result is intentionally more interpretive than Theorems 1 and 2. It does not characterise the implemented weighting rule directly; instead it formalises a limitation principle for posterior predictors built from an explicit self-model $\tilde{\pi}$ of the policy being updated.

Notation. Throughout this section: π_θ denotes the true policy at checkpoint θ; $\tilde{\pi}_\theta$ denotes the optimizer's self-model of π_θ; and

$$\hat{\pi}_{q_T} \;:=\; \mathbb{E}_{\theta' \sim q_T}[\tilde{\pi}_{\theta'}]$$

is the policy induced by the Meta-SWAG posterior predictive.

Theorem 1.3 (Conceptual self-knowledge bound under an explicit self-model)**.** *Let π_θ, $\tilde{\pi}_\theta$, and $\hat{\pi}_{q_T}$ be as above. Define the self-knowledge gap*

$$L_{\text{self}}(\theta) \;:=\; D_{\text{KL}}\Big(\pi_\theta \,\Big\|\, \tilde{\pi}_\theta\Big)\,,$$

and suppose the log-ratio of the self-model to the posterior predictive is uniformly bounded:

$$C \;:=\; \sup_x \; \log \frac{\tilde{\pi}_\theta(x)}{\hat{\pi}_{q_T}(x)} \;<\; \infty. \tag{3}$$

Then

$$D_{\mathrm{KL}}\Big(\pi_\theta \,\Big\|\, \hat{\pi}_{q_T}\Big) \;\le\; L_{\mathrm{self}}(\theta) \;+\; C \;+\; O(T^{-1}), \tag{4}$$

where the $O(T^{-1})$ term reflects finite-sample approximation error in the Meta-SWAG moment estimates under a geometric-mixing rate assumption on the checkpoint trajectory.

Correction: Error 1 (notation, eq. (??)): The original wrote $\pi_{q_T}(x)$ in the denominator (no hat), inconsistent with the definition of $\hat{\pi}_{q_T}$ introduced two lines earlier. Corrected to $\hat{\pi}_{q_T}(x)$.

Error 2 (the form of the bound, eq. (12) in the original): The original stated

$$D_{\mathrm{KL}}(\pi_\theta \| \hat{\pi}_{q_T}) \;\le\; \tfrac{1+C^2}{2}\, L_{\mathrm{self}}(\theta) + O(T^{-1}).$$

This *multiplicative* form in L_{self} does not follow from the stated proof strategy. The standard KL decomposition gives

$$D_{\mathrm{KL}}(\pi_\theta \| \hat{\pi}_{q_T}) = L_{\mathrm{self}}(\theta) + \mathbb{E}_{\pi_\theta}\left[\log \frac{\tilde{\pi}_\theta(x)}{\hat{\pi}_{q_T}(x)}\right].$$

The cross term is bounded by C *pointwise* (from condition (**??**)), yielding an *additive* bound $L_{\mathrm{self}} + C$. Young's inequality applied to a product $C\sqrt{L_{\mathrm{self}}}$ would produce terms of the form $\alpha L_{\mathrm{self}} + C^2/(4\alpha)$ — still additive in C^2, not a multiplicative factor on L_{self}. A multiplicative bound in L_{self} alone would require an explicit relationship between C and L_{self}, which is not assumed. The corrected bound (**??**) is additive, as the proof demands.

Error 3 ($O(T^{-1})$ rate unjustified): The original asserted this rate without stating the mixing-rate or smoothness assumptions that imply it. The corrected statement makes the geometric-mixing hypothesis explicit as a precondition.

Error 4 (proof sketch interleaved with Figure 2 caption): The original embedded the Figure 2 placeholder caption mid-proof (a PDF column-extraction artefact). The proof sketch and figure caption are separated below.

Proof sketch. Decompose the KL divergence by inserting $\tilde{\pi}_\theta$:

$$D_{\mathrm{KL}}(\pi_\theta \| \hat{\pi}_{q_T}) \;=\; \underbrace{D_{\mathrm{KL}}(\pi_\theta \| \tilde{\pi}_\theta)}_{L_{\mathrm{self}}(\theta)} \;+\; \underbrace{\mathbb{E}_{\pi_\theta}\left[\log \frac{\tilde{\pi}_\theta(x)}{\hat{\pi}_{q_T}(x)}\right]}_{\text{cross term}}.$$

Since $\log(\tilde{\pi}_\theta(x)/\hat{\pi}_{q_T}(x)) \le C$ pointwise by condition (**??**), taking expectation under π_θ gives the cross term $\le C$. Adding the $O(T^{-1})$ finite-sample error from the Meta-SWAG moment approximation (which converges at this rate under geometric mixing of the checkpoint trajectory) yields bound (**??**). □

Remark 1.3.1 (Figure placeholder). Figure 2 (placeholder) is intended to show matrix-game validation with two panels: (left) empirical variance ratio versus the predicted HM/AM ratio for point estimate, uniformly weighted SWAG, and Meta-SWAG variants; (right) posterior top-eigenvalue ratio versus basin-width estimate across games and initialisation regimes. This figure validates Theorems **??** and **??** empirically and does not form part of the proofs.

Corollary 1.3.1 (Alignment under self-knowledge bound)**.** *Let π_{θ_0} be the base LLM policy (i.e. π_θ evaluated at a reference pre-alignment checkpoint θ_0), let $\tilde{\pi}$ be the aligner's internal model of π_{θ_0}, and let $\hat{\pi}_{q_T}$ be the aligned posterior predictive policy. Then alignment quality,*

*measured as $D_{\mathrm{KL}}(\pi_{\theta_0} \| \hat{\pi}_{q_T})$, is upper-bounded by the aligner's self-knowledge gap $L_{\mathrm{self}}(\theta_0) = D_{\mathrm{KL}}(\pi_{\theta_0} \| \tilde{\pi})$ plus the uniform log-ratio C of (**??**):*

$$D_{\mathrm{KL}}(\pi_{\theta_0} \| \hat{\pi}_{q_T}) \;\leq\; L_{\mathrm{self}}(\theta_0) + C + O(T^{-1}).$$

In particular, posterior averaging cannot compensate arbitrarily for a poor internal model of the policy being aligned.

Correction: Error 5 (notation conflict): The original Corollary 3 introduced a bare π for the "base LLM policy" without relating it to π_θ from Theorem 3, creating a notational gap. The corrected version identifies $\pi = \pi_{\theta_0}$ at an explicit reference checkpoint θ_0, resolving the ambiguity.

Remark 1.3.2 (Practical force of the self-knowledge bound). Theorem **??** identifies a limitation principle for adaptive systems: better uncertainty estimates cannot outrun a poor internal model of the policy being updated. Its practical force depends on whether L_{self} and C can be estimated or bounded in real aligners. The current framework does not yet supply such estimates for DPO, PPO, or GRPO; this is an open empirical question.